Title of the Notes

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Just practice problems for Topology, Commutative Algebra, and perhaps Machine Learning and Advanced Algorithms.

1 Exam style stuff

I have to practice for MAT224 Commutative Algebra oral examiniation so these problems are more about intuition, allowing less rigor.

Problem 1.1 (Large algebra identification problem)**.** Let

$$R = k[x, y],\ I = (x^2, xy, y^2),\ A = R/I.$$

What can you say about A? In particular, is it Noetherian or Artinian? Can you give a k-basis or normal form for its elements?
Compute $\sqrt{I}$. Then describe $V(I) \subseteq k^2$. What information about I is visible from $V(I)$?
Is I prime? Is I primary? Give the quickest convincing argument you can.
Determine the maximal ideals of A. Is A local? If so, identify the maximal ideal $\mathfrak{m}$, and compute enough powers of $\mathfrak{m}$ to see what happens.
Compute $\mathrm{Ann}_A(\bar{x})$ and $\mathrm{Ann}_A(\bar{y})$. Are either of these prime ideals?
Consider mutliplication by $\bar{x}$,

$$\mu_{\bar{x}} : A \to A,\ a \mapsto \bar{x}a.$$

Without being given a sequence, identify $\ker \mu_{\bar{x}}$, $\mathrm{im}\, \mu_{\bar{x}}$, $\mathrm{coker}\, \mu_{\bar{x}}$.

Solution. We have $I = (x, y)^2$. We can then say that the ring is Noetherian by using Hilbert's Basis Theorem twice $k \to k[x] \to k[x][y] = k[x, y]$ with Noetherian preserved. Then $I \subseteq R$ is Noetherian so $R/I = A$ is Noetherian.

Artinian does not hold for R since $(x) \supsetneq (x^2) \supsetneq (x^4) \supsetneq \cdots$ never stabilizes. We have an exact sequence

$$0 \to (x, y)(x, y) \to R = k[x, y] \to A \to 0.$$

We have that R is Artinian if and only if $(x, y)^2$ and A are Artinian. We know that R is not Artinian, and $(x, y)^2$ is not Artinian either so we can't conclude anything about A using this.

What we can notice is that any polynomials with x degree ≥ 2 vanish, same for y, while terms in both x and y vanish also. Thus all polynomials have only constant terms, terms in x and terms in y. Thus A is an R-algebra, more specifically a k-vector space with basis $\{1, \bar{x}, \bar{y}\}$.

The root $\sqrt{I}$ is (x, y). We obviously have $(x, y) \subseteq \sqrt{I}$. For $ax + by \in (x, y)$ we have

$$(ax + by)^2 = a^2x^2 + (2ab)xy + b^2y^2$$

which is in I. Now, all generators of I are divisible by by either x or y so $I \subseteq (x, y)$. We have (x, y) prime (and even maximal) so $\sqrt{I} \subseteq (x, y)$ (since $f^n \in I \subseteq J$ implies $f \in J$ due to primality).

We do not have that I is prime, since x^2 is in I, but x is not. It is, however, maximal. We have already established that (x, y) is maximal and $I = (x, y)^2$ so

$$(x, y)^2 \subseteq I \subseteq (x, y).$$

Thus I is (x, y)-primal.

The maximal ideals of A map one to one with the maximal ideals of R containing I.

As mentioned we have $(x, y) \supseteq (x^2, xy, y^2)$ and (x, y) is maximal so

$$(\bar{x}, \bar{y}) = \frac{(x, y)}{(x^2, xy, y^2)}.$$

We can use the third isomorphism theorem to prove that $A/(\bar{x}, \bar{y}) \cong k$.

For uniqueness, suppose $f \in A$. Then $f = a + bx + cy$. The ideal $\mathfrak{m} = (\bar{x}, \bar{y})$ consists of all elements where the constant term is zero, $a = 0$, so an element f not in $(\bar{x}, \bar{y})$ is just $f = a \in k \setminus \{0\}$, which is invertible since k is a field.

As we glimpsed earlier, $\mathfrak{m}^2 = (\bar{x}, \bar{y})(\bar{x}, \bar{y}) = (\bar{x}^2, \bar{x}\bar{y}, \bar{y}^2) = 0$.

We have

$$\text{Ann}_A(\bar{x}) = \{a \in A \mid a\bar{x} = 0.\}$$

Equivalently, in $A = R/I$ we have that the annihilator of $\bar{x}$ are all $a = \alpha + \beta\bar{x} + \delta\bar{y}$ such that

$$a\bar{x} = \alpha\bar{x} + \underbrace{\beta\bar{x}^2}_{0} + \underbrace{\delta\bar{y}\bar{x}}_{0} = 0.$$

We have that $\bar{x}$ is a basiselement of the space, requiring $\alpha = 0$. β, δ can be any elements at all so

$$a = 0 + \beta\bar{x} + \delta\bar{y} \in (\bar{x}, \bar{y}) = \mathfrak{m}.$$

It is not terribly surprising that $\text{Ann}_A(\bar{x})$ is contained in $\mathfrak{m}$, since it is maximal, but actually $\text{Ann}_A(\bar{x}) = \mathfrak{m}$.

We can see this since, for $a\bar{x} + b\bar{y} \in (\bar{x}, \bar{y})$ we have

$$(a\bar{x} + b\bar{y})\bar{x} = a\bar{x}^2 + b\bar{y}^2 = 0.$$

Thus $\mathrm{Ann}_A(\bar{x}) = \mathfrak{m}$, the maximal ideal.

Now, the exact same holds for $\bar{y}$, for the same reasons, and it is not hard to see.

Now for multiplication at $\bar{x}$ we get a natural short exact sequence

$$0 \to A \xrightarrow{\mu_{\bar{x}}} A \xrightarrow{\pi} \mathrm{Ann}_A(\bar{x}) \to 0$$

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Problem (Tensor products and quotient modules)**.** Let $R = k[x, y]$, $I = (x^2, xy)$, and $J = (y)$. Put $M = R/I$ and $N = R/J$.
Consider the R-linear map

$$\begin{aligned} \varphi : R &\longrightarrow M \otimes_R N, \\ r &\longmapsto (r + I) \otimes (1 + J). \end{aligned}$$

a) Prove that φ is surjective.
b) Prove that $I + J \subseteq \ker \varphi$.
c) Determine whether the induced map

$$\bar{\varphi} : R/(I + J) \longrightarrow M \otimes_R N$$

is an isomorphism.
d) Determine $M \otimes_R N$ explicitly and exhibit a k-vector-space basis.

Proof of a). Take $\bar{m} \otimes \bar{n} \in M \otimes_R N$. Then there is at least one each of m_0, n_0 such that $\bar{m} = m_0 + I$ and $\bar{n} = n_0 + J$.

Take $m_0 n_0 \in R$. Then

$$m_0 n_0 \mapsto (m_0 n_0) + I \otimes 1 + J = m_0 + I \otimes n_0 + J = \bar{m} \otimes \bar{n}.$$

■

Proof of b). Let $f \in I + J$. Then

$$f = ax^2 + bxy + cy, \ a, b, c \in k[x, y].$$

Then

$$f \mapsto (ax^2 + bxy + cy) + I \otimes 1 + J.$$

Then

$$(\underbrace{ax^2 + bxy}_{\bar{0}} + cy) + I \otimes 1 + J = cy + I \otimes 1 + J = I \otimes J = 0.$$

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Proof of c). Intuitively, φ is surjective with $\ker\varphi = I + J$ (I think) so we know, by the first isomorphism theorem, that

$$R/(I+J) \cong M \otimes_R N.$$

The induced map $\bar{\varphi}$ sends $\bar{r} = r + (I+J)$ to

$$\bar{r} \mapsto \bar{r} + I \otimes 1 + J.$$

As mentioned, $\ker\varphi = I + J$ which corresponds to 0 in $R/(I+J)$. The induced map retains surjectivity as we simply take

$$\bar{m}\bar{n} \mapsto \bar{m} \otimes \bar{n}.$$

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Solution to d). We have

$$M \otimes_R N \cong R/(I+J) = R/((x^2, xy) + (y)).$$

We have

$$(x^2, xy) + (y) = (x^2, xy, y)$$

so

$$R/(I+J) \cong k[x]/(x^2).$$

We then have a basis $\{1, x\}$ for $k[x]/(x^2)$ which corresponds to the basis

$$\{\bar{1}, \bar{x}\}.$$

■

Extra.

1.

Let

$$\begin{aligned} \beta : (R/I) \times (R/J) &\longrightarrow R/(I+J), \\ (r+I, s+J) &\longmapsto rs + (I+J). \end{aligned}$$

We show that β is well-defined.

Suppose $\bar{r} = r_0 + I = r_1 + I$ with $r_0 \neq r_1$.

Then

$$\begin{aligned}\beta(r_0 + I, s + J) &= (r_0 s) + (I + J) \\ &= \overline{r_0 s} \\ &= \bar{r_0}\bar{s} \\ &= \bar{r_1}\bar{s} \\ &= (r_1 s) + (I + J) \\ &= \beta(r_1 + I, s + J).\end{aligned}$$

The point is that

$$\bar{x}\bar{y} = \overline{xy}.$$

The argument is analogous for $\bar{s}$.

2.

For bilinearity, fix first s, supressing the equivalence class notation for now.

Then

$$\begin{aligned}\beta(r_0(r_1 + r_2), s) &= ((r_0(r_1 + r_2))s) \\ &= r_0(r_0 s + r_1 s) \\ &= r_0(\beta(r_0, s) + \beta(r_1, s)).\end{aligned}$$

We could also be more rigorous than this. The argument for fixed r is the same.

Since β is an R-bilinear function we invoke the tensor products unviersal property, giving a map $\hat{\beta} : R/I \otimes_R R/J \to R/(I + J)$ such that $\beta \circ ((r, s) \mapsto r \otimes s) = \hat{\beta}$. ■

Problem (Compactness and separation axioms)**.** Let X be a compact Hausdorff space.

a) Let $x \in X$, and let $F \subseteq X$ be closed with $x \notin F$. Prove that there exist disjoint open sets $U, V \subseteq X$ such that

$$x \in U, \qquad F \subseteq V.$$

b) Prove that every compact Hausdorff space is normal. That is, if $A, B \subseteq X$ are disjoint closed sets, there exist disjoint open sets $U, V \subseteq X$ with

$$A \subseteq U, \qquad B \subseteq V.$$

c) Let X be an infinite set equipped with the cofinite topology

$$\mathcal{T} = \{\varnothing\} \cup \{U \subseteq X : X \setminus U \text{ is finite}\}.$$

Determine, with proofs, whether X is compact, T_1, Hausdorff, and normal.

Proof of a). This follows from compact Hausdorff $\Rightarrow$ normal. $\{x\}$ is closed, since X is Hausdorff, so since X is normal, we can separate

$$\{x\},\ F$$

into disjoint open sets containing them.

Below will be a proof from first principles.

For $y \in F$ we can, by Hausdorffness find

$$U_y \ni x$$

and

$$V_y \ni y$$

disjoint and open.

We have

$$F \subseteq \bigcup_{y \in F} V_y.$$

Since X is compact we have

$$F \subseteq \bigcup_{i=1}^{n} V_{y_i}.$$

Set

$$V = \bigcup_{i=1}^{n} V_{y_i}$$

is open. For x, let U be

$$U = \bigcap_{i=1}^{n} U_{y_i}.$$

This is a finite intersection of open sets so open.

Now

$$U \cap V = \bigcup_{j=1}^{n} \left(\left(\bigcap_{i=1}^{n} U_{y_i} \right) \cap V_{y_j} \right).$$

We have

$$\left(\bigcap_{i=1}^{n} U_{y_i} \right) \cap V_{y_j} \subseteq U_{y_j} \cap V_{y_j}$$

so each term is empty.

Thus $U \cap V = \emptyset$.

■

Proof of (b). We proceed as above. If X is Hausdorff it is T_1.

For fixed $x \in A$ construct

$$V_x = \bigcup_{j=1}^{n} V_{y_i}$$

and

$$U_x = \bigcap_{i=1}^{n} U_{y_i}$$

as in (a) with the property that $U_x \cap V_x$

We do this for all $x \in A$. Then

$$\bigcup_{x \in A} U_x$$

covers A and can be reduced to

$$\bigcup_{i=1}^{n} U_{x_i}.$$

Corresponding V_{x_i}'s independently cover B, so their intersection also does, and it is disjoint to each U_{x_i} and thus disjoint to their union.

■

Proof of (c). It is indeed T_1 since

$$\{x\} = X \setminus (X \setminus \{x\})$$

meaning

$$X \setminus \{x\}$$

is open.

Take two distinct $x, y \in X$. Suppose $x \in U,\ y \in V$ are disjoint open sets.

Then

$$|X \setminus U| = n,\ |X \setminus V| = m$$

for nonzero integers m, n.

Since $U \cap V = \emptyset$ we have

$$U^c \cup V^c = X.$$

Then

$$X \cap (U^c \cup V^c) = X,$$

but then

$$|X| = |X \cap (U^c \cup V^c)| = |X \cap (U^c) \cup X \cap (V^c)| \leq n + m,$$

giving

$$n + m \geq |X| \geq \aleph_0,$$

a clear contradiction.

Thus X is not Hausdorff.

Let $\mathcal{U} = \{U_\alpha\}$ be a cover of some U. Pick some $U_1 \in \mathcal{U}$. Then $X \setminus U_1 = \{x_1, \dots, x_n\}$. For each x_i, if $x_i \in U$, take one of the U_α which contains it. Repeat until all x_i are accounted for. Then we have a cover with at most $n+1$ elements.

The space is not Hausdorff and thus not normal. Suppse it was normal. We know the space is T_1 so $\{x\}, \{y\}$ are closed. Then we can partition them into open disjoint sets, implying Hausdorffness. No good. ■

Problem. Let

$$A = k[x]/(x^2).$$

a) Prove that every element $a + b\bar{x}$ with $a \neq 0$ is invertible. Construct the inverse explicitly.
b) Deduce that every zero divisor is nilpotent.
c) Recall that an ideal Q is primary if and only if every divisor in R/Q is nilpotent. Use this to conclude that

$$(x^2, y) \subseteq k[x, y]$$

is primary.

Proof of (a/b). We see that

$$(a + b\bar{x})(a - b\bar{x}) = a^2 + b^2\bar{x}^2 = a^2.$$

a is nonzero, so a^2 is nonzero in k, making it invertible.

Taking

$$(a + b\bar{x})(a - b\bar{x})a^{-2}$$

we get

$$a^2a^{-2} = 1.$$

Thus, for nonzero a, the inverse is $(a + b\bar{x})^{-1} = a^{-2}(a - b\bar{x})$.

Suppose $a + b\bar{x} \notin (x^2)$ is nonzero with

$$(a + b\bar{x})c = \bar{x}^2.$$

Then, clearly $\bar{x} \mid c$ so $\bar{x}^2 | c^2$. ■

Proof of (c). From (b) we have that all zero divisor in $k[x]/(x^2) \cong k[x, y]/(x^2, y)$ is nilpotent, making $Q = (x^2, y)$ primary. ■